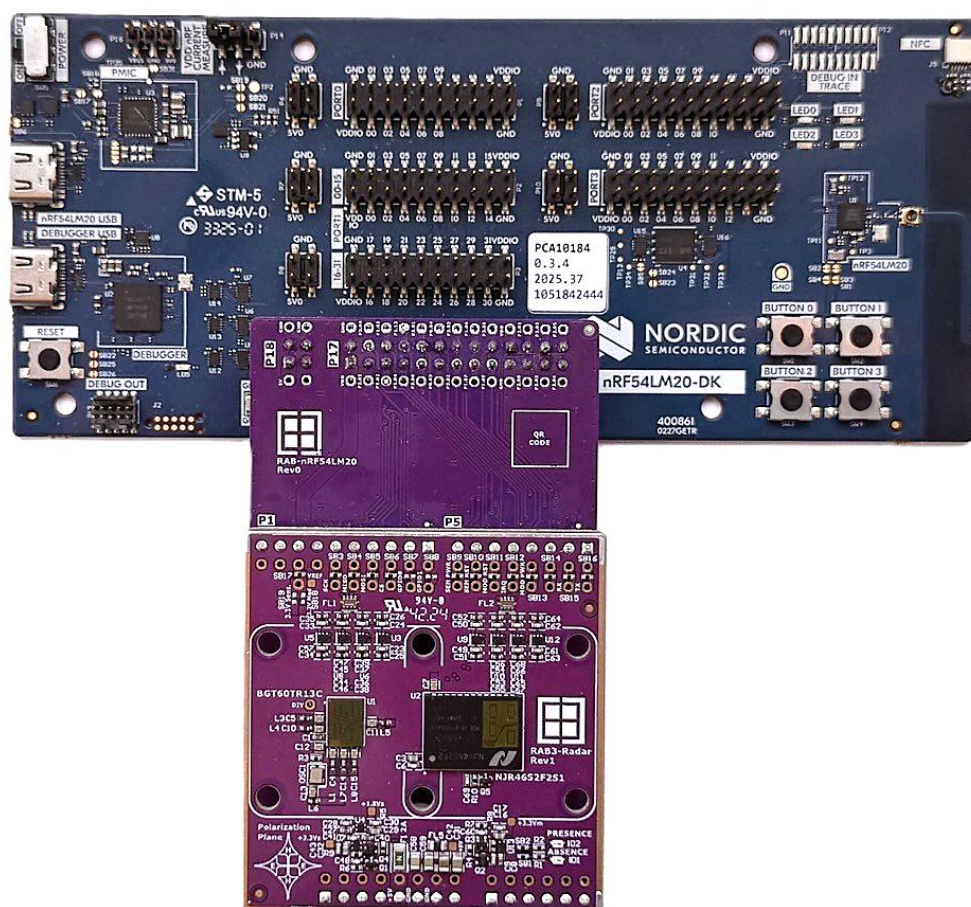


# Using Nordic nRF54LM20 DK and Infineon BGT60TR13C for presence detection

## Application note



## Versions

| Version    | Date            | Rationale                   |
|------------|-----------------|-----------------------------|
| <b>0.1</b> | July 13, 2026   | Draft. Authors: DC1         |
| <b>1.0</b> | October 9, 2026 | First release. Authors: KOA |

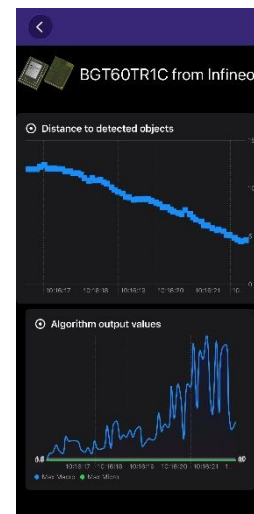
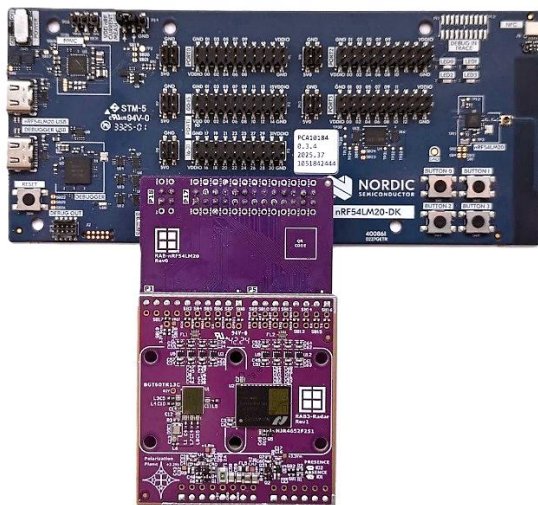
## Legal Disclaimer

The evaluation board is for testing purposes only and, because it has limited functions and limited resilience, is not suitable for permanent use under real conditions. If the evaluation board is nevertheless used under real conditions, this is done at one's responsibility;  
any liability of Rutronik is insofar excluded.

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## Introduction



The solution enables to detect presence using the Infineon BGT60TR13C radar sensor. The Nordic nRF54LM20-DK board collects the measurement data from the BGT60TR13C and process them. The results of the algorithm are sent over BLE to the iOS and Android app.

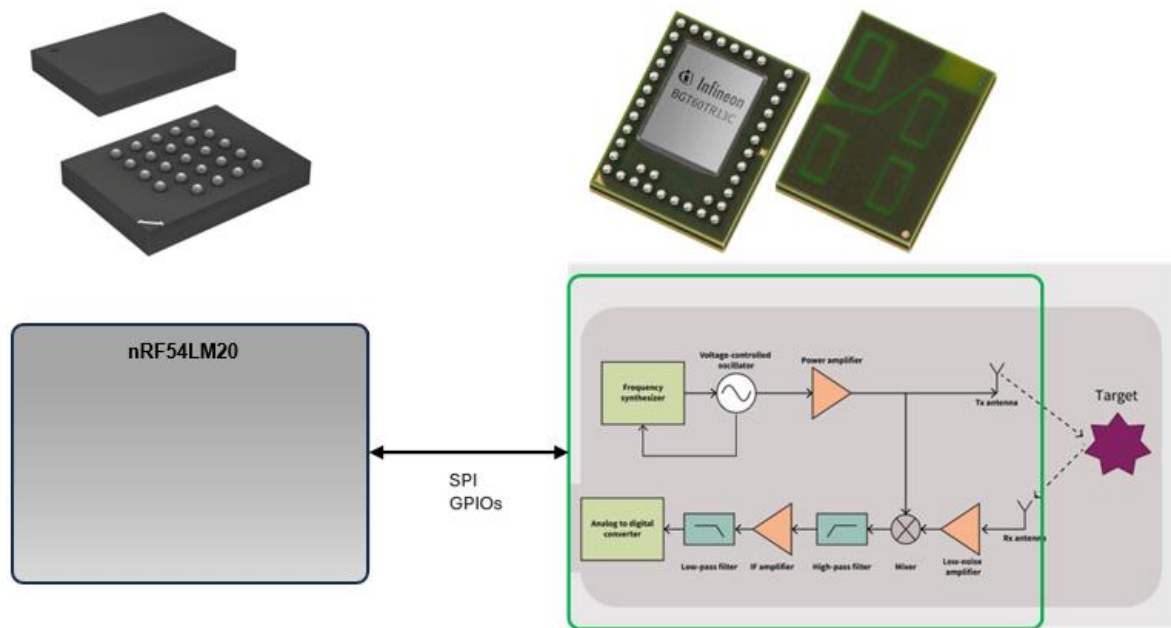
## Hardware

### Required boards

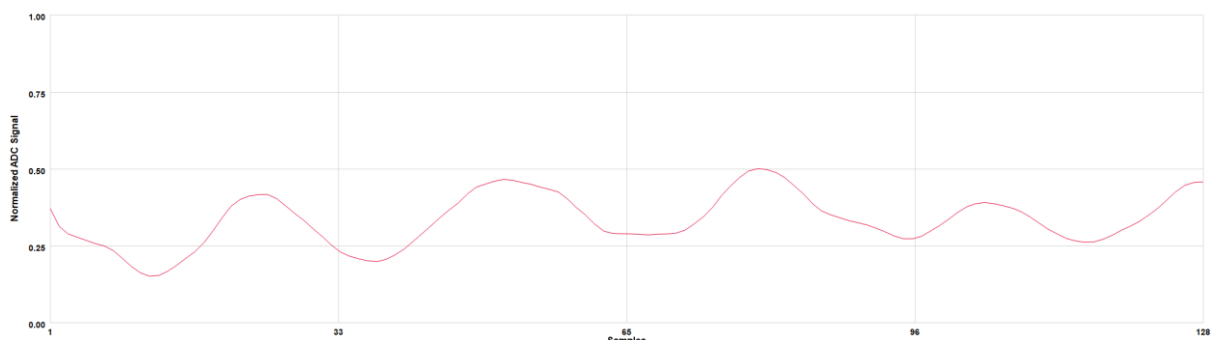
- Nordic nRF54LM20-DK development kit.
- RAB3-Radar.
- RAB-nRF54LM20 (adapter board enabling to connect the RAB3-Radar to the nRF54LM20-DK).

### Radar sensor data flow

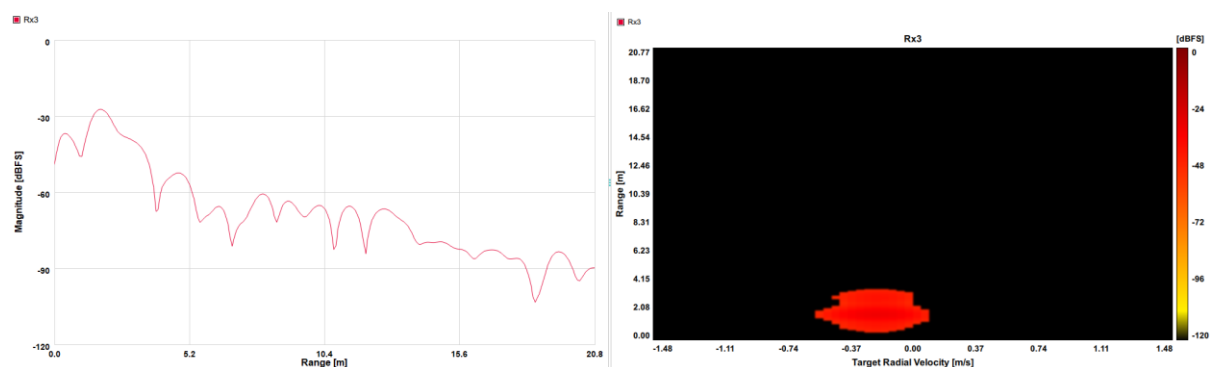
The BGT60TR13C embeds antennas and signal conditioning block. It delivers raw measurements to the microcontroller over SPI. The maximum SPI communication frequency supported by the radar sensor is 50 MHz. In our example with the Nordic nRF54LM20 DK, the maximum stable SPI communication frequency verified in this project is 8 MHz.



The values sent by the BGT60TR13C to the microcontroller are ADC samples that look like:



Using digital signal processing, it is possible to extract information from those raw values, like a fact of presence for example. This is the role of the microcontroller.



## Software

### Overview

The firmware is available on [GitHub](#) and runs on the Nordic nRF54LM20 DK. The project is opened, built and flashed using Visual Studio Code with the nRF Connect for VS Code extension.

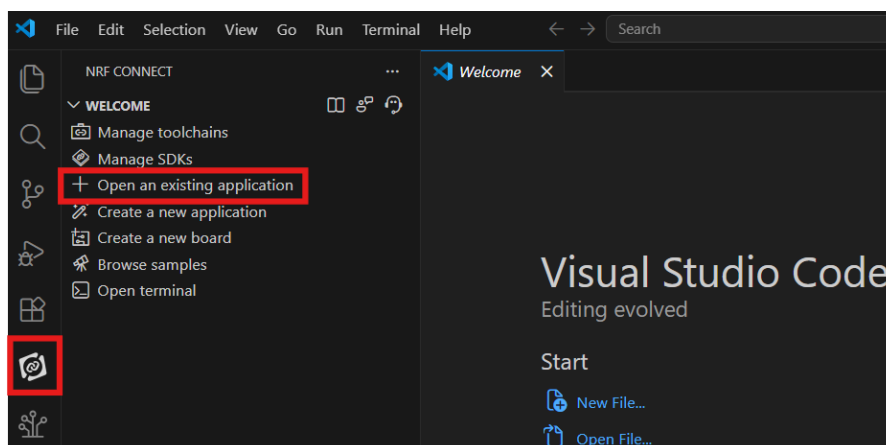
Before opening the project, the nRF Connect SDK, the corresponding toolchain and the nRF Connect extension for Visual Studio Code have to be installed.

```
SDK:          nRF Connect SDK v3.3.0
Toolchain:    nRF Connect SDK Toolchain v3.3.0
```

Nordic provides an official tutorial for the installation and basic workflow:

[https://www.nordicsemi.com/Products/Development-tools/nRF-Connect-for-VS-Code/Tutorials?utm\\_source=chatgpt.com](https://www.nordicsemi.com/Products/Development-tools/nRF-Connect-for-VS-Code/Tutorials?utm_source=chatgpt.com)

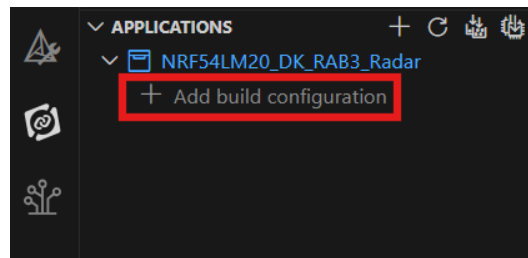
### Import project in VSC



Open **Visual Studio Code** and select the **nRF Connect** extension on the left side. In the **Welcome** section, click on **Open an existing application**. Select the project folder of the radar firmware. After selecting the folder, the project is added to the **Applications** section of the nRF Connect extension.

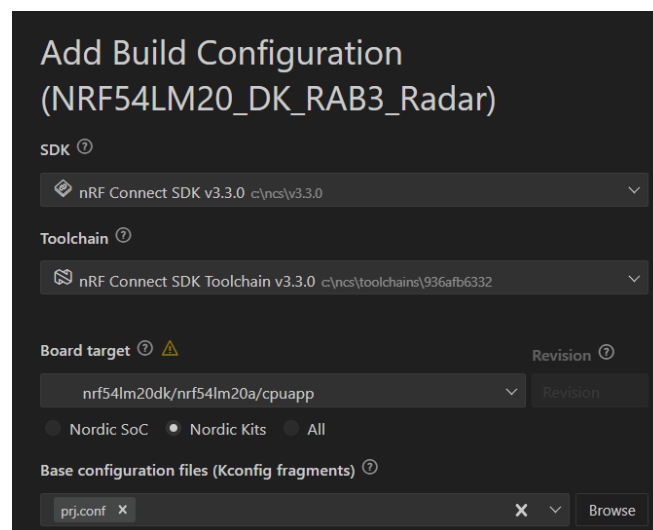
### Add build configuration

After the project has been added to the Applications section, click on **Add build configuration**.



In the build configuration window, select **nRF Connect SDK v3.3.0** and **nRF Connect SDK Toolchain v3.3.0**.

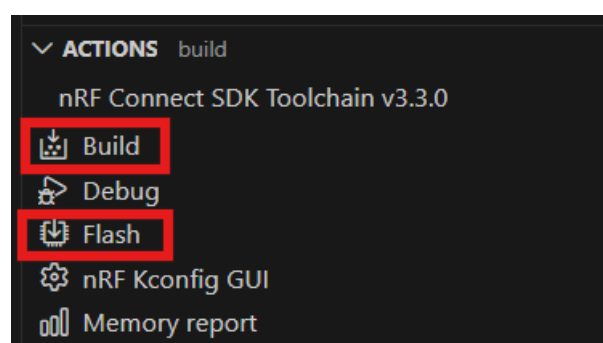
For this project, the following board target is used: **nrf54lm20dk/nrf54lm20a/cpuapp**  
The base configuration file is **prj.conf**.



Click on **Generate and Build** to generate the build folder and build the firmware for the selected target.

## Build and flash the firmware

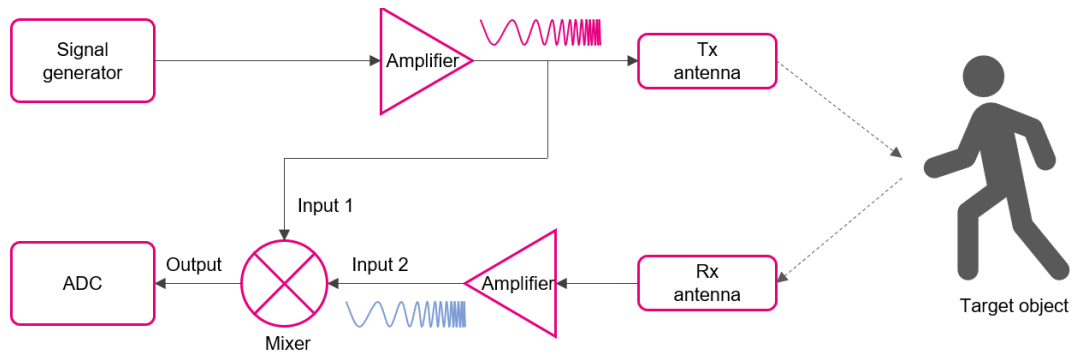
After the build has finished successfully, the build configuration appears in the **Actions** section. Click on **Build** to rebuild the firmware after code changes. Click on **Flash** to program the firmware to the connected nRF54LM20 DK.



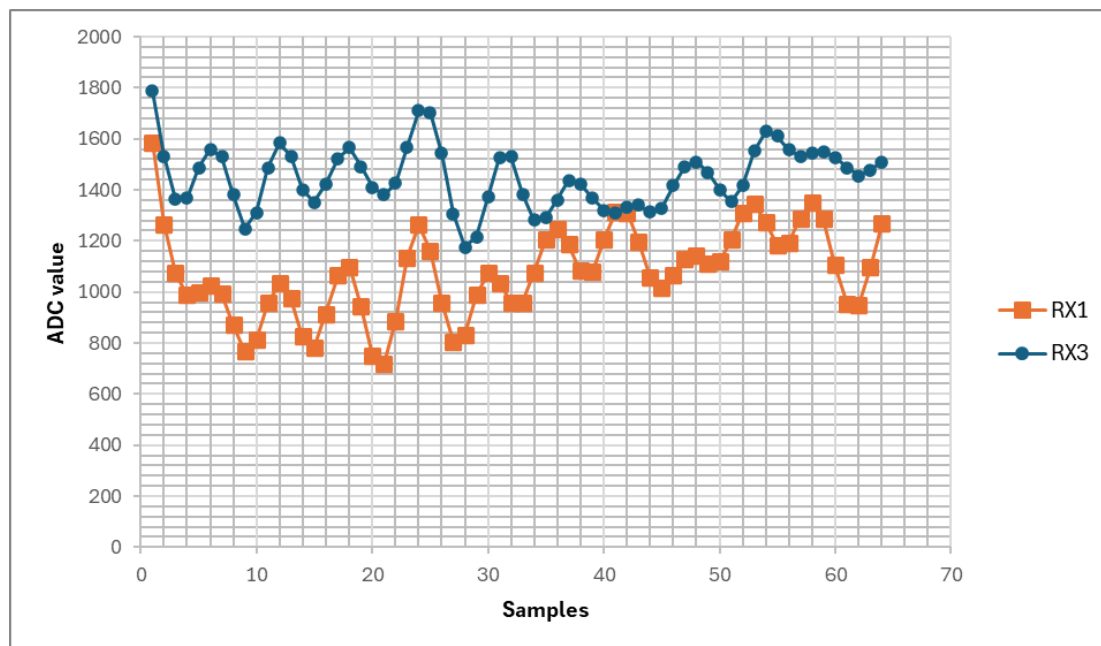
## Calculations

### Raw values from radar sensor

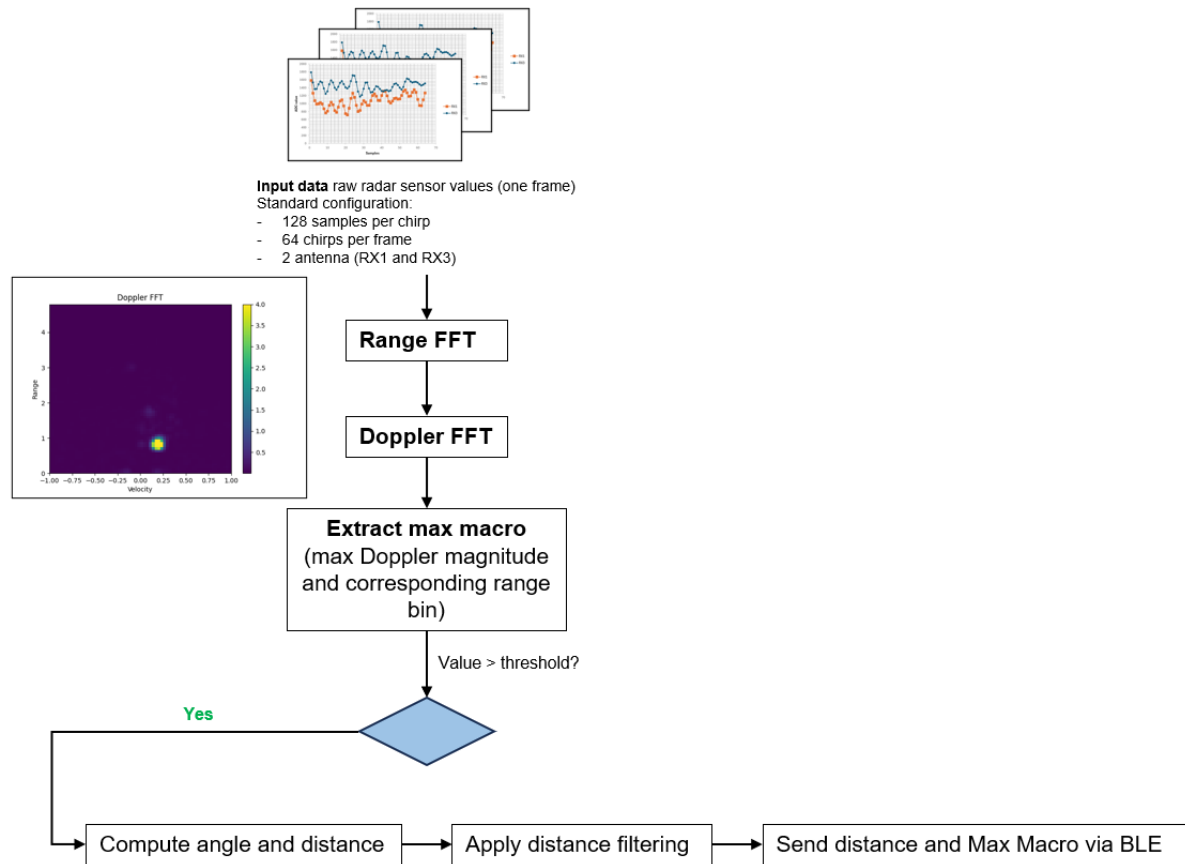
The BGT60TR13C delivers raw values (sampled by the internal ADC) over SPI bus.



One packet of raw values, also named frame, is composed by multiple chirps. Each chirp contains X (in our case 64) samples per antenna (as below).



## Algorithm



If the Max Macro value is above the defined threshold, the detected range bin is used to compute the displayed distance value. Before the value is sent to the smartphone application, a distance filter is applied to reduce sudden jumps.